

Steps to make the transformation real

To begin, a factory's current state must be assessed across procurement, inventory, production, supply chain, maintenance, and quality. Capturing data, analysing it to find performance gaps, and identifying improvement opportunities is critical. Decisions must be justified with a clear return on investment (ROI).

Once the groundwork is laid, the focus shifts to process-level transformation. Every process, from manufacturing and assembly to quality inspection and material handling, must be simulated and analysed. Simulation tools, ranging from Level 0 to Level 5, ensure layouts and flows are optimised. Both static and dynamic simulations play roles depending on applicability. This includes simulating human-dependent activities to ensure expected output.

Whether greenfield or brown-field, factory setups must be virtually validated. Physical and virtual validation ensures design, space, equipment, and processes align with production goals. A communication network, such as 5G, Wi-Fi, or wired internet, must be selected to ensure seamless machine-to-machine interaction.

The goal is to create a comprehensive, integrated ecosystem that connects manufacturing systems, factory operations, and decision-making capabilities.

Operator training is equally vital. Experienced workers may know the process, but new operators must quickly learn to avoid quality errors and achieve required performance. Tools such as digital work instructions, AR/VR, and interactive platforms enable just-in-time training, even without formal classrooms.

On the execution side, MES platforms are central. They manage production orders, track workflows and OEE (overall equipment

effectiveness), handle quality and maintenance, and monitor energy usage. Real-time dashboards provide visibility into factory operations.

For older machines, especially in brownfield settings, retrofitting is essential. Sensors can convert even 60- or 70-year-old machines into data-generating units. Newer machines with built-in communication must be integrated using suitable protocols such as MQTT, Modbus, ProfiBus, ProfiNet, and OPC-UA. Once connected, this data is then fed into AI-powered systems for predictive maintenance and quality control.

This brings us to the latest approach—6D Smart Manufacturing Transformation, which includes six pillars:

- 1. Factory scanning.** Validating that physical construction aligns with digital designs
- 2. Change alerts.** Flagging early deviations for timely corrective actions
- 3. Cost anchoring.** Tracking how costs accumulate during physical and digital implementation
- 4. Environmental monitoring.** Leveraging data for environmental responsibility and compliance
- 5. Sustainability.** Aligning with India's national agenda for sustainable manufacturing
- 6. AI integration.** Using AI to manage complexity, drive insights, and support continuous improvement

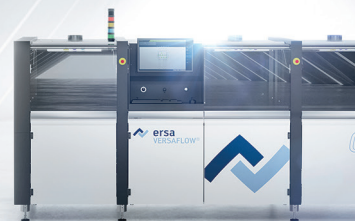
Industries are moving forward in their manufacturing transformation journey. This evolution extends beyond digitisation, aiming to create intelligent, agile, and sustainable ecosystems that address the changing needs of automotive, aerospace, and electronics sectors. **EFY**

Based on a session titled 'Manufacturing Transformation Vs Electronic Industry' delivered by Prashant Kumkar, Practice Head Smart Manufacturing, Tata Technologies Ltd, at the EFY Expo held at the Auto Cluster Exhibition Center, Pune, on 16 May 2025. Transcribed and curated by Shubha Mitra, Assistant Editor at EFY.

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